



Xi'an Jiaotong-Liverpool University

西交利物浦大学

INT305 Machine Learning

Lecture 5

Neural Network and Back Propagation

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Neural network

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

Neuralnetwork

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(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$

Neuralnetwork

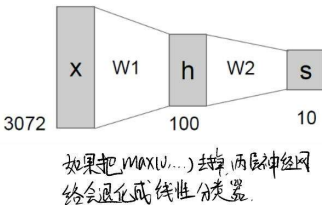
第一步: $W_1 x$
第一部分和先前的线性函数一模一样! 这里的 W_1 是第一个权重矩阵。它把输入的 x (3072 个像素值) 转换成一个中间结果。根据图示, 这个中间结果是一个有 100 个数值的向量 h 。我们可以把这 100 个数值看作是网络从原始像素中学到的“特征”或者“模式”。它们不再是原始的像素了, 而是更高层次的组合, 比如“可能存在曲线的边缘”、“可能有一块蓝色的区域”等等。

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2 \max(0, W_1 x)$

第三步: W_2
经过 ReLU 激活函数处理后, 我们得到了一个包含了非负特征的向量 h 。
现在, 我们再用第二个权重矩阵 W_2 对这个特征向量 h (100 个值) 进行一次线性变换, 最终得到 10 个类别的分数 s 。
这一步就相当于, 模型根据前面学到的“曲线边缘”、“蓝色区域”等特征, 来做最终的判断: “嗯, 根据这些特征来看, 这张图有 90% 的可能 是猫, 10% 的可能 是狗。”



第一层($W_1 + \text{ReLU}$): 从原始像素中
提取有用的、抽象的特征。
第二层(W_2): 根据这些提取出来的
特征, 进行最终分类。

第二步: $\max(0, \dots)$
这是它与线性函数最根本的区别! 这个函数叫做
激活函数 (Activation Function), 而 $\max(0, z)$ 是
其中最常用的一种, 它的名字叫 ReLU (Rectified
Linear Unit), 中文叫“修正线性单元”。
它的工作原理超级简单: 如果一个数是正数, 就
保持不变; 如果一个数是负数, 就把它变成 0。
这是为了引入非线性 (non-linearity)。这就好比我们
给了模型一个“开关”, 让它能够学会更复杂的
、弯弯曲曲的决策边界。没有这个激活函数,
无论我们叠加多少层线性函数, 最终结果都等价
于一个线性函数, 就像把几个直尺首尾相连, 得
到的还是一根更长的直尺。但有了 ReLU 这个“关
节”, 我们就可以把直尺折叠, 去拟合更复杂的形
状!

最内层: $\max(0, W_1x)$
和之前一样, 这是我们的第一个隐藏层。它接收原始的图片像素 x , 通过 W_1 计算, 再经过 ReLU 激活, 输出了第一组比较初级的特征 (比如一些边缘、颜色块等)。

Neuralnetwork

Neural Network: without the brain stuff

(Before) Linear score function: $f = Wx$

(Now) 2-layer Neural Network: $f = W_2\max(0, W_1x)$

or 3-layer Neural Network:
 $f = W_3\max(0, W_2\max(0, W_1x))$

中间层: $\max(0, W_2...)$
现在是关键的一步! 它不再直接把这些初级特征用于最终分类, 而是把它们作为输入, 再进行一次相同的操作: 乘以第二个权重矩阵 W_2 , 然后再次通过 ReLU 激活函数。
这一层就是第二个隐藏层。它的作用是把第一层学到的简单特征 (边缘、色块) 进行组合, 从而学习到更复杂、更有意义的特征 (比如“眼睛的轮廓”、“鸟的喙”)。

最外层: $W_3(...)$
这是我们的输出层。它接收第二个隐藏层输出的那些复杂的组合特征, 然后乘以 W_3 , 最终计算出每个类别的最终分数 f 。

每增加一个“隐藏层”, 网络就获得了从上一层特征中学习更高级、更抽象特征的能力。这就是“深度学习”中“深度 (Deep)”的来源——我们通过堆叠更多的层来构建一个更深的网络

Activation functions

它可以把任何输入的数值都“压缩”到 0 和 1 之间。这非常有用, 比如在需要输出一个概率值 (概率总是在 0 到 1 之间) 的场景下。

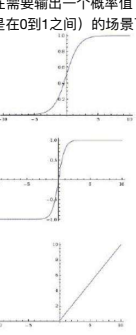
Sigmoid
 $\sigma(x) = \frac{1}{1 + e^{-x}}$
和 Sigmoid 很像, 但它把数值“压缩”到 -1 和 1 之间。它的输出是以 0 为中心的, 这一点比 Sigmoid 更好, 通常能让模型收敛得更快。

ReLU
 $\max(0, x)$
计算超级快, 而且在正数区域不会有梯度消失的问题, 所以学习得很快。

有个叫“Dying ReLU”的问题。如果一个神经元的输入总是负数, 它就会一直输出 0, 梯度也永远是 0, 这个神经元就再也无法学习了

可以把激活函数想象成每个神经元的“性格开关”或者“决策模型”。它决定了接收到信号后, 这个神经元应该被“激活”到什么程度, 以及应该向下一层传递什么样的信息。它的核心使命, 我们再强调一遍, 就是引入非线性, 让网络能够学习复杂的数据模式。

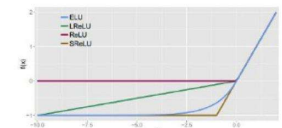
Leaky ReLU
当输入是负数时, 它不再输出 0, 而是输出一个很小的值 (比如 $0.1x$)。这就好比给“死掉”的神经元做了一点心肺复苏, 让它还有机会被重新激活。



LeakyReLU
 $\max(0.1x, x)$

Maxout $\max(w_1^Tx + b_1, w_2^Tx + b_2)$

ELU
 $f(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha(\exp(x) - 1) & \text{if } x \leq 0 \end{cases}$

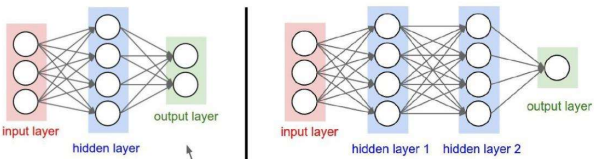


Maxout 这是一个更广义的激活函数, 它本身是可以学习的。可以看作是 ReLU 和 Leaky ReLU 的泛化版本, 效果很好, 但计算开销和参数数量也更大。

ELU 是 Leaky ReLU 的一个更平滑的升级版, 既解决了“Dying ReLU”问题, 又具有接近 0 均值的输出。但计算会比 ReLU 复杂一点。

Neuralnetwork

Neural Networks: Architectures



"2-layer Neural Net", or
"1-hidden-layer Neural Net"

一个隐藏层，一个输出层，

"3-layer Neural Net", or
"2-hidden-layer Neural Net"

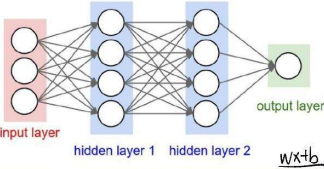
两个隐藏层，一个输出层

"Fully-connected" layers

每层之间的节点，必须是完全连接。
前一层的所有神经元，都与后一层所有神经元相连。

Neuralnetwork

Example Feed-forward computation of a Neural Network



输入数据
第二个隐藏层

```
# forward-pass of a 3-layer neural network:  
f = lambda x: 1.0/(1.0 + np.exp(-x)) # activation function (use sigmoid)  
x = np.random.randn(3, 1) # random input vector of three numbers (3x1)  
h1 = f(np.dot(W1, x) + b1) # calculate first hidden layer activations (4x1)  
h2 = f(np.dot(W2, h1) + b2) # calculate second hidden layer activations (4x1)  
out = np.dot(W3, h2) + b3 # output neuron (1x1)
```

sigmoid 激活函数
→ 第一个隐藏层

输出层

★ Gradient Descent 梯度下降

Where we are...

$$s = f(x; W) = Wx$$

$$L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

$$L = \frac{1}{N} \sum_{i=1}^N L_i + \sum_k W_k^2$$

want $\nabla_W L$

梯度指向了让损失上升最快的方向, 那我们要考虑怎么更新权重 w 来降低损失, 让模型变得更好。

所以梯度下降就是沿着梯度的反方向走

scores function 对于一个给定的输入, 模型对每个可能类别有多大的信心

SVM loss (hinge loss) 用来衡量模型在单个样本上预测得有多差, 值越高, 说明模型在这一类别上犯的错误越大

data loss + regularization

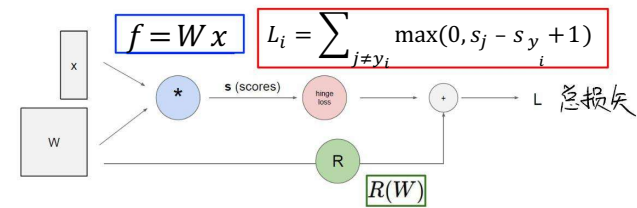
对模型在整个数据集上表现的最终评价

$\sum_{i=1}^N L_i$: Data loss (数据损失) 把所有的单个样本损失 (L_i) 加起来, 再求一个平均值, 代表了我们的模型在整个训练集上的平均不满意程度。

$\sum W_k^2$: 惩罚过于复杂的模型。把所有权重 w 的平方加起来, 如果权重太大, 这一项就会很大, 导致总损失变大。主要用于防止过拟合。

★ Computational Graph

Computational Graph



ComputationalGraph

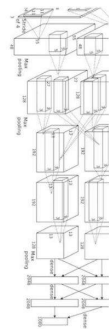
Convolutional Network
(AlexNet)

卷积神经网络

input image

weights

loss



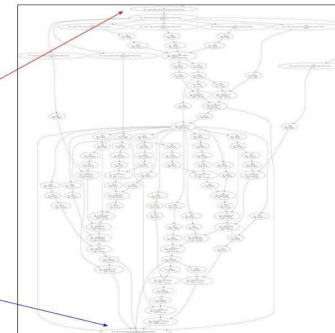
ComputationalGraph

Neural Turing Machine

神经网络图灵机

input tape

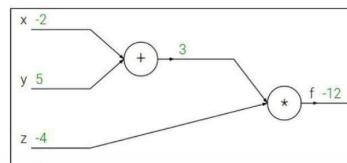
loss



Example1

$$f(x,y,z) = (x+y)z$$

e.g. $x = -2, y = 5, z = -4$



Example1

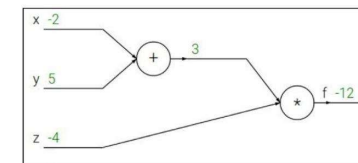
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$$f = qz \quad \frac{\partial f}{\partial q} = z, \frac{\partial f}{\partial z} = q$$

Want: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$



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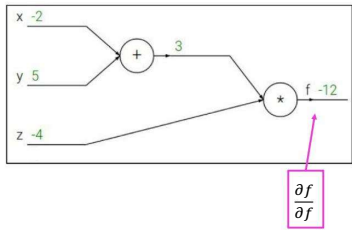
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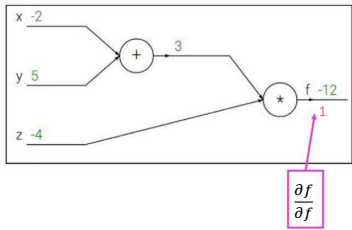
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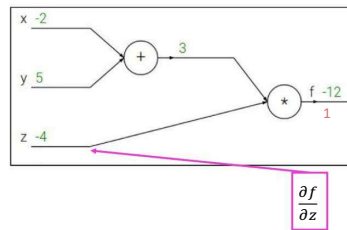
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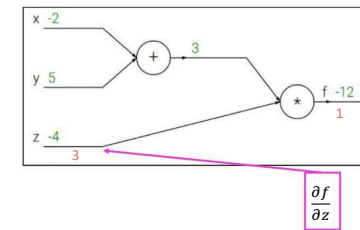
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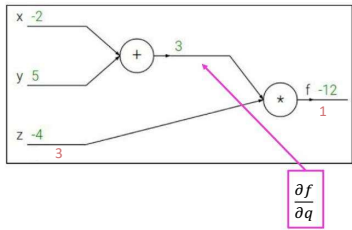
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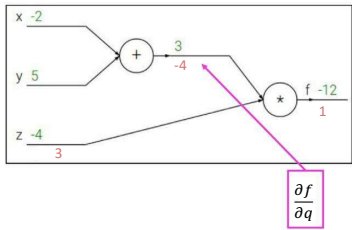
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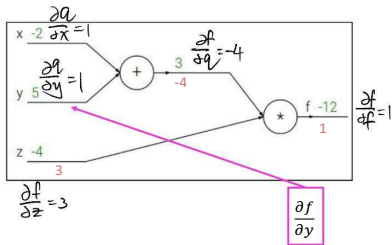
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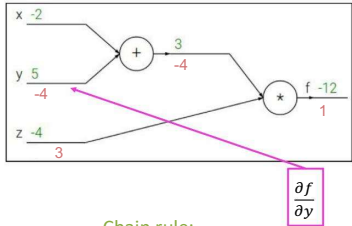
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Chain rule:

$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$

$(-4) \cdot 1 = -4$

Example1

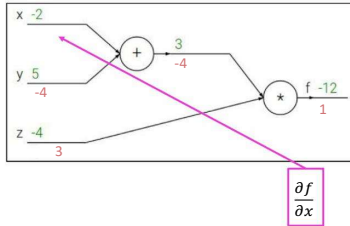
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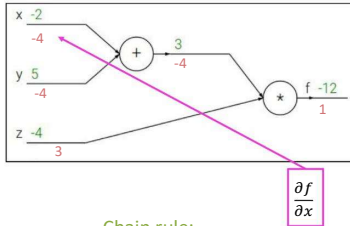
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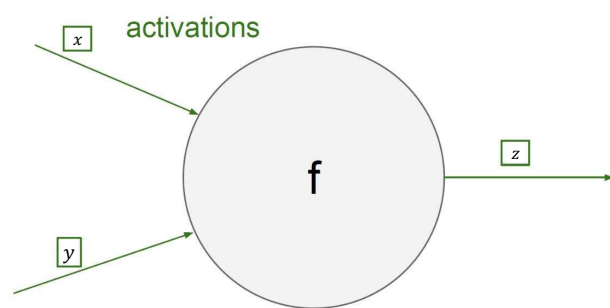
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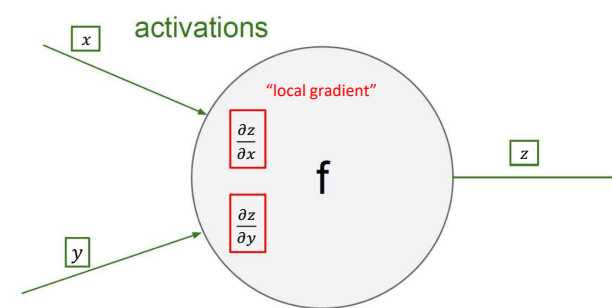


Chain rule:
 $\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial y}$ *for y?*
 $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \frac{\partial q}{\partial x}$

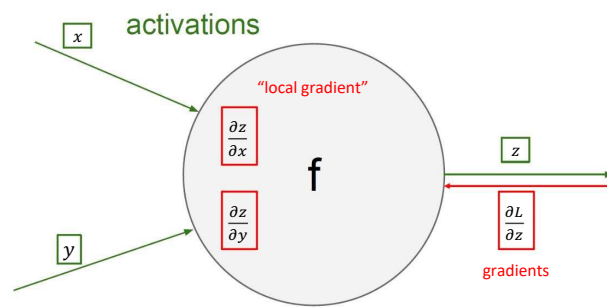
Chainrule



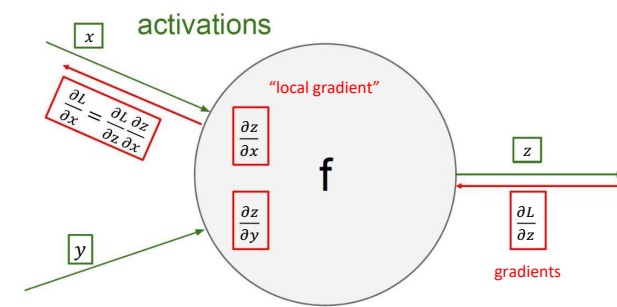
Chainrule



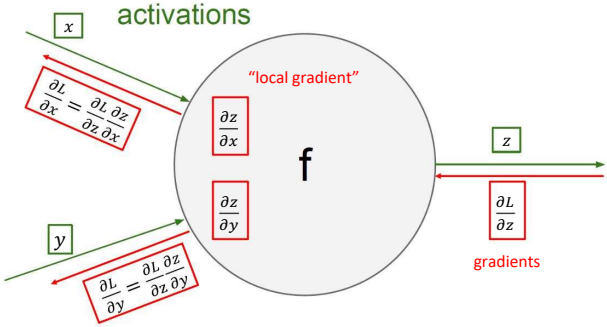
Chainrule



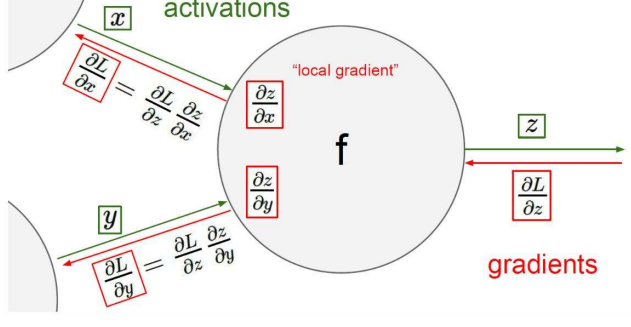
Chainrule



Chainrule

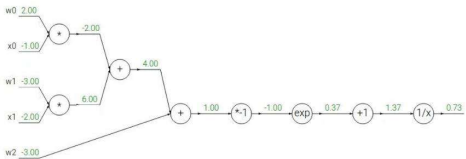


Chainrule



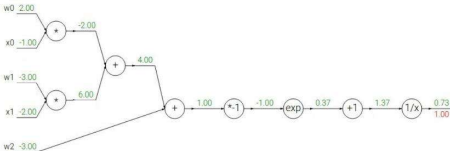
Chainrule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



Chainrule

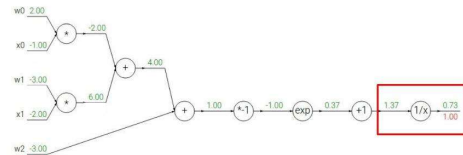
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$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$	$\left \right.$	$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$	$\left \right.$	$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chainrule

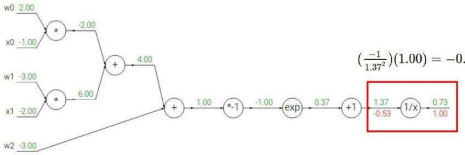
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$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$	$\left \begin{array}{l} f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2 \\ f_c(x) = c + x \rightarrow \frac{df}{dx} = 1 \end{array} \right.$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$	

Chainrule

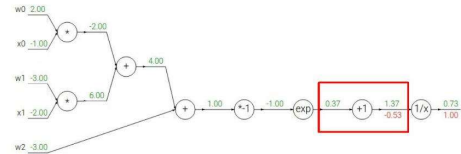
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$	$\left \begin{array}{l} f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2 \\ f_c(x) = c + x \rightarrow \frac{df}{dx} = 1 \end{array} \right.$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$	

Chainrule

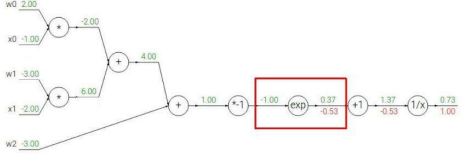
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$	$\left \right.$	$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$	$\left \right.$	$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chainrule

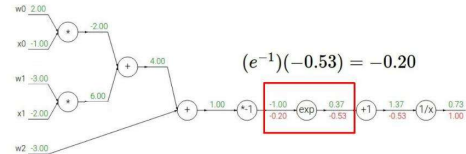
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



$f(x) = e^x$	$\rightarrow$	$\frac{df}{dx} = e^x$	$\left \right.$	$f(x) = \frac{1}{x}$	$\rightarrow$	$\frac{df}{dx} = -1/x^2$
$f_a(x) = ax$	$\rightarrow$	$\frac{df}{dx} = a$	$\left \right.$	$f_c(x) = c + x$	$\rightarrow$	$\frac{df}{dx} = 1$

Chainrule

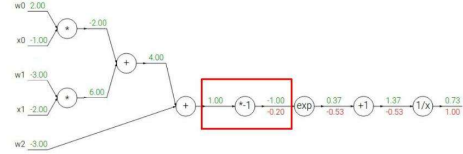
Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$



$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$	$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$
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Chainrule

Another example: $f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$

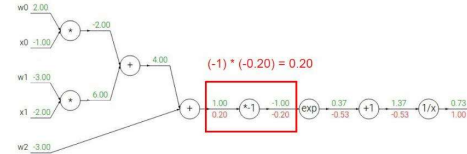


$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$	$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$
$f_a(x) = ax \rightarrow \frac{df}{dx} = a$	$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$

Chainrule

Another example:

$$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2 x_2)}}$$

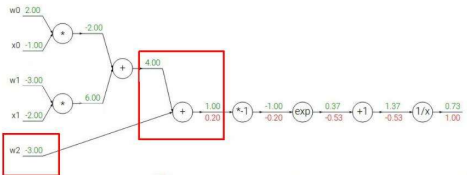


$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$	$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$
$f_a(x) = ax \rightarrow \frac{df}{dx} = a$	$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$

Chainrule

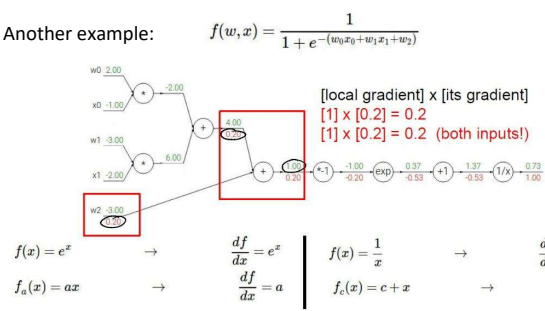
Another example:

$$f(w, x) = \frac{1}{1 + e^{-(w_0 x_0 + w_1 x_1 + w_2 x_2)}}$$

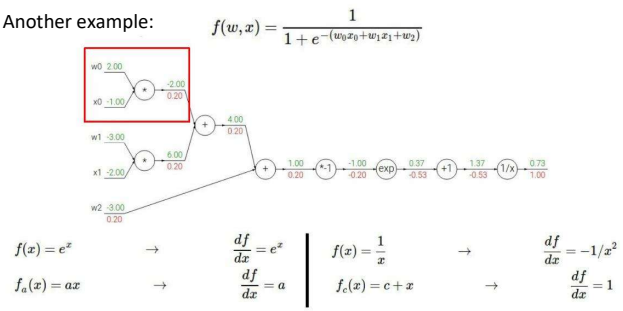


$f(x) = e^x \rightarrow \frac{df}{dx} = e^x$	$f(x) = \frac{1}{x} \rightarrow \frac{df}{dx} = -1/x^2$
$f_a(x) = ax \rightarrow \frac{df}{dx} = a$	$f_c(x) = c + x \rightarrow \frac{df}{dx} = 1$

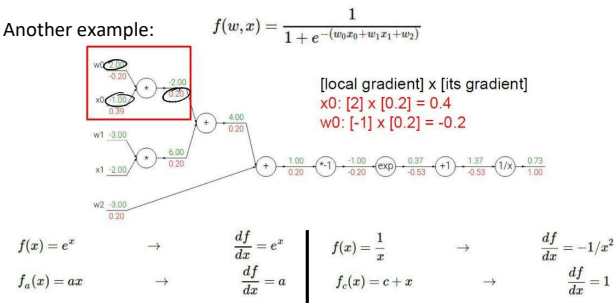
Chainrule



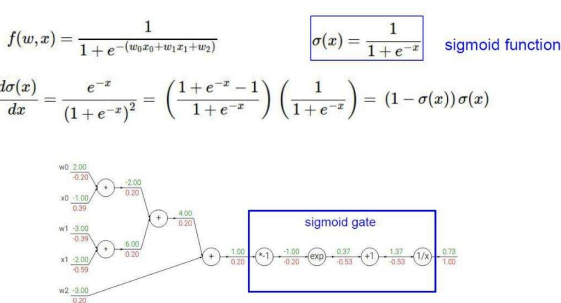
Chainrule



Chainrule



Sigmoid

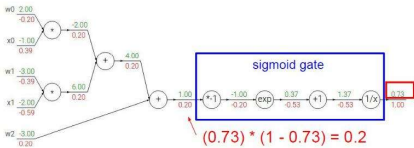


Sigmoid

$$f(w, x) = \frac{1}{1 + e^{-(w_0x_0 + w_1x_1 + w_2x_2)}}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$
 sigmoid function

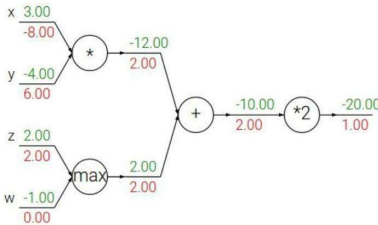
$$\frac{d\sigma(x)}{dx} = \frac{e^{-x}}{(1 + e^{-x})^2} = \left(\frac{1 + e^{-x} - 1}{1 + e^{-x}}\right) \left(\frac{1}{1 + e^{-x}}\right) = (1 - \sigma(x))\sigma(x)$$



Patterninbackwardflow

Patterns in backward flow

- addgate:** gradient distributor
- maxgate:** gradient router
- mulgate:** gradient... “switcher”?



Exercise 1

Pooling units take n values x_i , $i \in [1, n]$ and compute a scalar output whose value is invariant to permutations of the inputs.

1. The Lp-pooling module takes positive inputs and computes

$$y = (\sum_i x_i^p)^{\frac{1}{p}}, \text{ assuming we know that } y' = \frac{\partial L}{\partial y}, \text{ what is } x'_i = \frac{\partial L}{\partial x_i}?$$

2. The log-average module computes $y = \frac{1}{\beta} \ln(\frac{1}{n} \sum_i \exp(\beta x_i))$,

assuming we know that $y' = \frac{\partial L}{\partial y}$, what is $x'_i = \frac{\partial L}{\partial x_i}$?

(1). $y = (\sum_{i=1}^n x_i^p)^{\frac{1}{p}}, q = \sum_{i=1}^n x_i^p, y = q^{\frac{1}{p}}$

(2). $y = \frac{1}{\beta} \ln(\frac{1}{n} \sum_{i=1}^n e^{\beta x_i}), y = \frac{1}{\beta} \ln q$

① $\frac{\partial q}{\partial x_i} = p \cdot x_i^{p-1}$

② $\frac{\partial y}{\partial q} = \frac{1}{p} \cdot q^{\frac{1}{p}-1}$

③ $\frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial q} \cdot \frac{\partial q}{\partial x_i} = y' \cdot \frac{1}{p} q^{\frac{1}{p}-1} \cdot p \cdot x_i^{p-1} = y' \cdot (\sum_{i=1}^n x_i^p)^{\frac{1}{p}-1} \cdot x_i^{p-1}$

$q = \frac{1}{n} \sum_{i=1}^n e^{\beta x_i}$

① $\frac{\partial q}{\partial x_i} = \frac{1}{n} \cdot \beta \cdot e^{\beta x_i}$

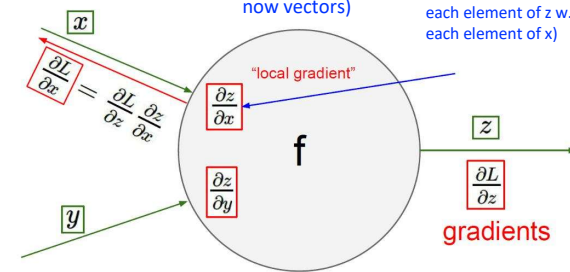
② $\frac{\partial y}{\partial q} = \frac{1}{\beta} \cdot \frac{1}{q}$

③ $\frac{\partial L}{\partial x_i} = \frac{\partial L}{\partial y} \cdot \frac{\partial y}{\partial q} \cdot \frac{\partial q}{\partial x_i} = y' \cdot \frac{1}{\beta} \cdot \frac{1}{q} \cdot \frac{1}{n} \cdot \beta \cdot e^{\beta x_i} = y' \cdot \frac{e^{\beta x_i}}{\sum_{i=1}^n e^{\beta x_i}}$

Gradients for vector

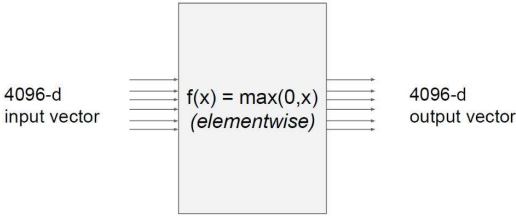
Gradients for vectorized code (x, y, z are now vectors)

This is now the Jacobian matrix (derivative of each element of z w.r.t. each element of x)



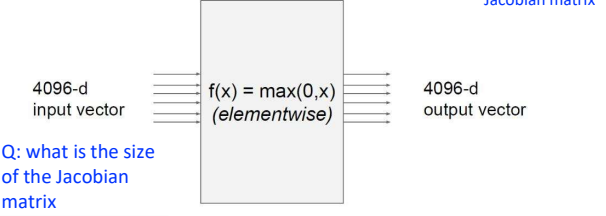
Gradientsforvector

Vectorized operation



Gradientsforvector

Vectorized operation

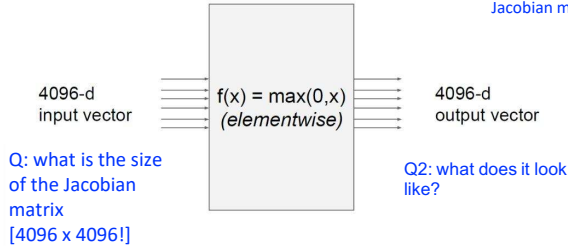


Gradientsforvector

Vectorized operation

$$\frac{\partial L}{\partial x} = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial L}{\partial f} \end{bmatrix}$$

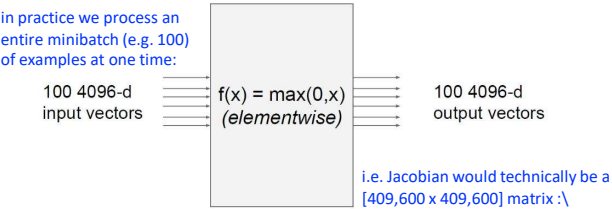
Jacobian matrix



Gradientsforvector

Vectorized operation

in practice we process an entire minibatch (e.g. 100) of examples at one time:



Gradientsforvector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

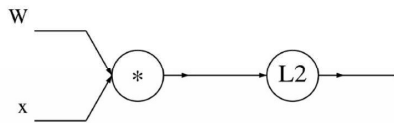
Gradientsforvector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$\Downarrow \Downarrow$
 $\in \mathbb{R}^n \in \mathbb{R}^{n \times n}$

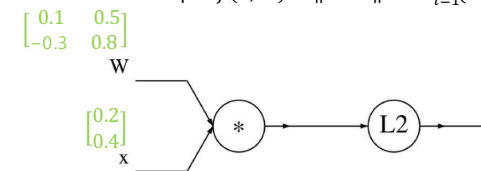
Gradientsforvector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



Gradientsforvector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



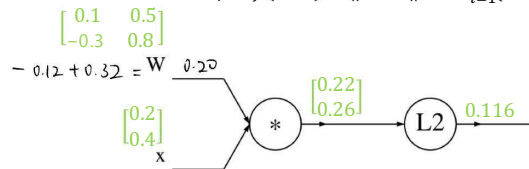
$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

Gradientsforvector

$$0.1 \times 0.2 + 0.5 \times 0.4 = 0.02 + 0.2 = 0.22$$

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$f(x, W) = \|W \cdot x\|^2 \quad q = Wx \quad f = q^T q \quad \text{Target: } \frac{\partial L}{\partial q}, \frac{\partial L}{\partial W}, \frac{\partial L}{\partial x}$$

① $\frac{\partial f}{\partial q} = \frac{\partial (q^T q)}{\partial q} = q + (q^T)^T = 2q = \begin{bmatrix} 2q_1 \\ 2q_2 \end{bmatrix}$ ② $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial x} = W^T \cdot 2q = \begin{bmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{bmatrix} \begin{bmatrix} 2q_1 \\ 2q_2 \end{bmatrix}$

Gradientsforvector

② $\frac{\partial f}{\partial W} = \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial W} = 2q \cdot x^T = \begin{bmatrix} 2q_1 \\ 2q_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} 2q_1 x_1 & 2q_1 x_2 \\ 2q_2 x_1 & 2q_2 x_2 \end{bmatrix}$

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x

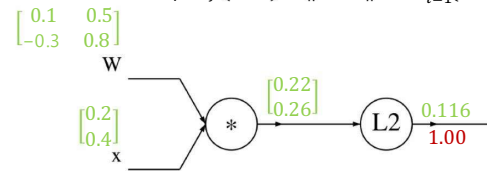


$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

Gradientsforvector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



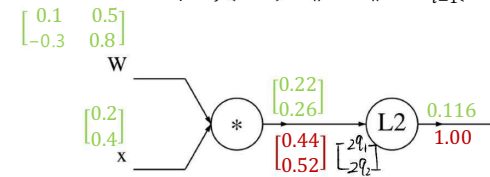
$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\star \begin{cases} \frac{\partial f}{\partial q_i} = 2q_i \\ \nabla_q f = 2q \end{cases}$$

Gradientsforvector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

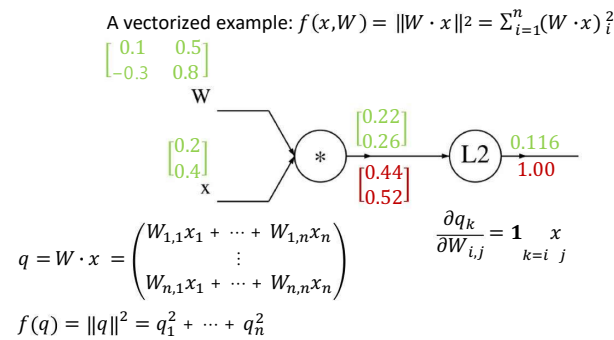


$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \dots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \dots + W_{n,n}x_n \end{pmatrix}$$

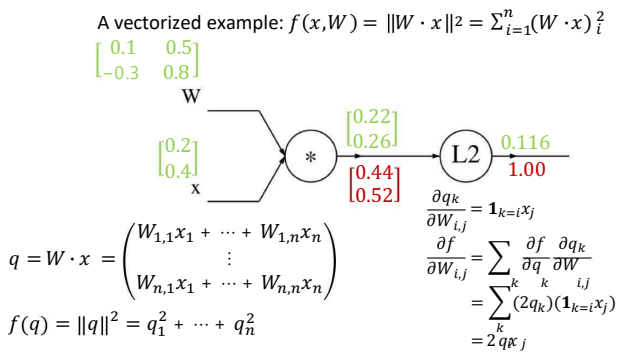
$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

$$\frac{\partial f}{\partial q_i} = 2q_i$$
$$\boxed{\nabla_q f = 2q}$$

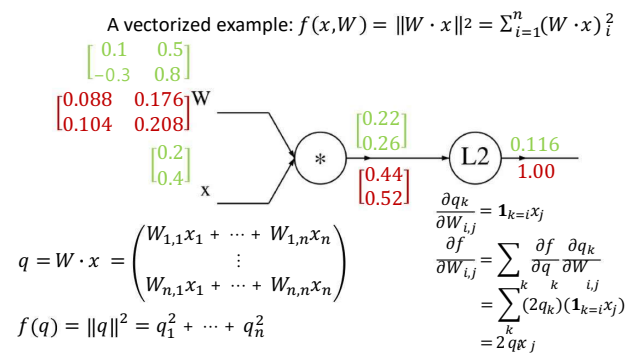
Gradientsforvector



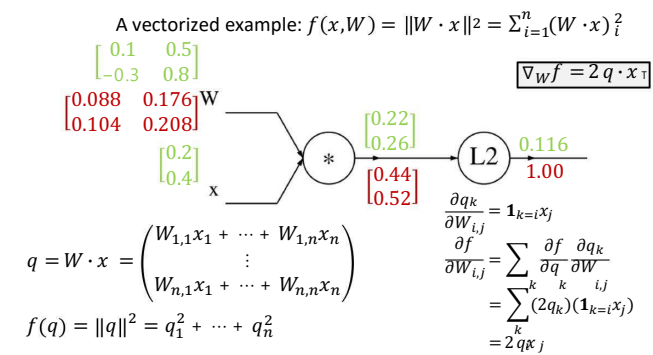
Gradientsforvector



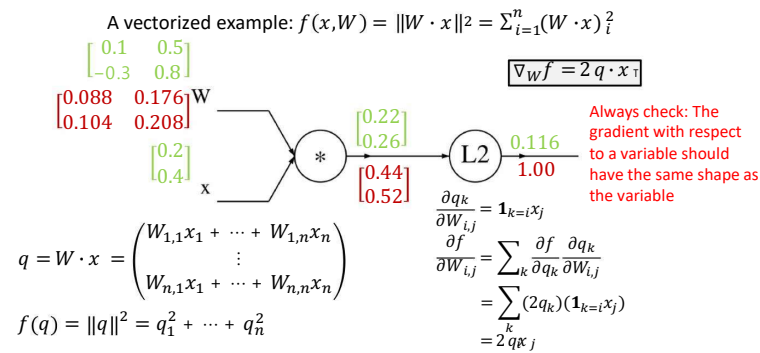
Gradientsforvector



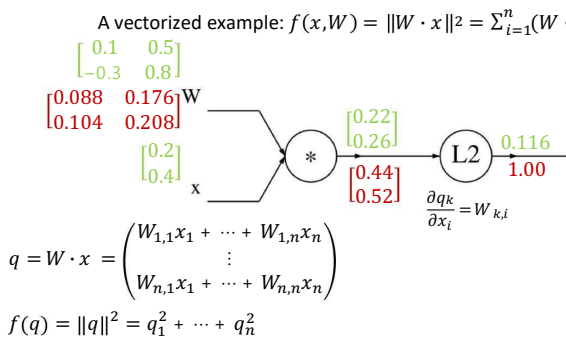
Gradientsforvector



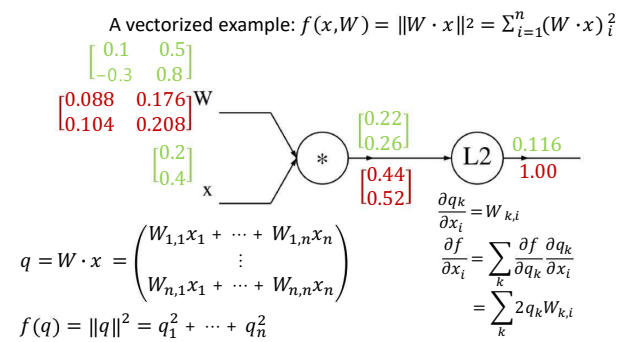
Gradientsforvector



Gradientsforvector



Gradientsforvector



Gradientsforvector

